

Chapter 8 Individual Assignment Report

Employee Directory with Vue 3, Axios, Express and MySQL

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1. Working User Interface

The application provides a single-page HR directory with an add/edit form, active headcount, server-backed search and sort controls, and a responsive employee table. Active and inactive employees are shown with status badges, and salaries are formatted in Malaysian Ringgit.

CHAPTER 8 INDIVIDUAL ASSIGNMENT

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Employee Directory

A single-page HR tool built with Vue 3, Axios, Express and MySQL.

NEW EMPLOYEE

Add Employee

Employee ID

EMP108

Full Name

Employee full name

Email

name@myhrpilot.test

Department

Select department

Position

Job title

Hire Date

mm/dd/yyyy

Salary (RM)

4500

☒ Currently active employee

Add Employee

DIRECTORY STATUS

6 active of 7 total

Search

Name, ID, email, or department

Sort by

Employee ID

Order

Ascending

Clear

EMPLOYEE RECORDS

Directory List

EMPLOYEE	DEPARTMENT	POSITION	HIRE DATE	SALARY	STATUS	ACTIONS
Lina Tan Mei Ling EMP101 - lina.tan@myhrpilot.test	HR	People Operations Executive	12 Apr 2021	RM 4,300.00	Active	<button>Edit</button> <button>Delete</button>
Daniel Lim Wei Jian EMP102 - daniel.lim@myhrpilot.test	IT	Systems Analyst	28 Sept 2020	RM 6,900.00	Active	<button>Edit</button> <button>Delete</button>
Farah Nadia Ismail EMP103 - farah.ismail@myhrpilot.test	Finance	Accounts Officer	07 Feb 2022	RM 5,100.00	Active	<button>Edit</button> <button>Delete</button>
Jason Goh Kai Sheng EMP104 - jason.goh@myhrpilot.test	Marketing	Brand Coordinator	19 Jun 2023	RM 3,900.00	Active	<button>Edit</button> <button>Delete</button>
Mira Aisyah Rahman EMP105 - mira.rahman@myhrpilot.test	Operations	Logistics Supervisor	04 Nov 2019	RM 6,200.00	Active	<button>Edit</button> <button>Delete</button>

Figure 1: Desktop layout showing the employee form, headcount, search/sort controls, and loaded records.

Additional Responsive Screenshot

The screenshot shows a tablet-width layout of the 'Employee Directory' application. At the top, a dark header contains the text 'CHAPTER 8 INDIVIDUAL ASSIGNMENT', 'Employee Directory', and 'A single-page HR tool built with Vue 3, Axios, Express and MySQL'. Below the header, a user profile card for 'Wong Jia Hui' with ID 'A24CS0312' is visible. The main content area is divided into two sections. The first section, titled 'NEW EMPLOYEE' and 'Add Employee', contains a form with fields for Employee ID (EMP108), Full Name (Employee full name), Email (name@myhrpilot.test), Department (Select department), Position (Job title), Hire Date (mm/dd/yyyy), and Salary (RM) (4500). There is a checkbox for 'Currently active employee' which is checked, and an 'Add Employee' button. The second section, titled 'DIRECTORY STATUS', shows '6 active of 7 total'. Below this is a search bar with the placeholder 'Name, ID, email, or department' and a 'Sort by' dropdown menu.

Figure 2: Tablet-width layout with controls stacked safely for narrower screens.

2. Learning Outcome Coverage

Learning Outcome	How the project satisfies it
LO1: Connect Vue to backend	App.vue loads employee records from the Express API and refreshes state after create, update, delete, search, and sort actions.
LO2: Use Axios	src/api/employeeApi.js defines one Axios instance with baseURL, timeout, JSON headers, request logging, response error mapping, and async helper functions.
LO3: Vue forms and validation	EmployeeForm.vue uses v-model.trim and v-model.number, validates required fields, shows inline errors, and emits save/cancel events to the parent.
LO4: CRUD with MySQL	server/index.js implements GET, POST, PUT, and DELETE routes using mysql2 prepared statements, plus server-side search and whitelisted sorting.

3. Technical Decisions

The project uses a single root package.json so the grader only needs to run npm install once. The backend stays inside server/, while Vue components are split into EmployeeForm, SearchSort, and EmployeeList. MySQL date values are returned as strings so hireDate remains stable as YYYY-MM-DD.

Prepared statements are used for all dynamic SQL values. Sorting is handled by mapping client sort keys to a server-side whitelist before building ORDER BY.

4. Challenges and Resolutions

Challenge	Resolution
Unsafe SQL sorting	Only known sort keys are mapped to fixed column names before ORDER BY is built.
Date consistency	The MySQL connection uses dateStrings: true so hireDate values do not shift timezone.
Clear validation	The client blocks invalid submits, and the backend returns structured field errors.
Tablet usability	Controls stack on narrow screens, and the employee table scrolls horizontally.

The application was verified with npm install and npm run build. Database setup instructions are included in README.md before running the final app.